Sri Lanka Institute of Information Technology

BSc (Hons) in Information Technology Specializing in Cyber Security

IT4010 - Research Project (Comprehensive Design and Analysis Project)

R26-CS-010

Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training

Progress Logbook

IT number	Student Name
IT22066916	WIJESINGHE R. D. P. T.

Date: 20th Feb - 14th Mar

Individual Component: Risk and Compliance Audit: Researching GDPR compliance requirements and planning the privacy attack testing framework for the AV data pipeline.

Summary of RP work carried out during the reporting period

Date	RP Work
20/02/2026	Received project briefing for R26-CS-010. Read through the project scope and understood the Risk and Compliance component requirements and deliverables.
21/02/2026	Read GDPR Article 5 (data minimisation) and Article 25 (privacy by design). Noted how each principle maps to the AV data pipeline design decisions.
24/02/2026	Studied Article 32 (security of processing). Understood how AES-256-GCM encryption satisfies its requirements for camera image storage in the pipeline.
25/02/2026	Reviewed pseudonymisation requirements under GDPR and confirmed that the CARLA simulator does not generate real PII, satisfying pseudonymisation by design.
26/02/2026	Started drafting the GDPR compliance requirements document. Identified five key GDPR principles applicable to the pipeline and documented them.
27/02/2026	Studied differential privacy concepts and the epsilon parameter to understand what the privacy budget auditing requirement means for the attack design.
28/02/2026	Completed first draft of the GDPR compliance requirements document and submitted for supervisor review.
03/03/2026	Incorporated supervisor feedback. Revised the pseudonymisation and encryption sections of the compliance document and finalised Phase 1 deliverable.
06/03/2026	Set up Python development environment. Installed required libraries including hmac, hashlib, cryptography, pandas and scipy for the attack scripts.
09/03/2026	Read research papers on HMAC-based certificate integrity verification to design the Attack 1 approach. Documented expected inputs, method and output.
11/03/2026	Studied the Opacus library documentation and differential privacy accounting methods to plan the Attack 2 privacy budget audit approach.
13/03/2026	Reviewed the Laplace mechanism and reconstruction attack literature to plan Attack 3. Documented the Laplace bound formula to be used.
14/03/2026	Studied Shokri et al. 2017 membership inference attack paper. Planned the shadow classifier approach for Attack 4 and documented the 0.55 AUC threshold.

Supervisor's comments.

Student progressing well in research project: Yes / No

If NO:

Project work to be carried out before next evaluation period:

Signature:

Date: 15/03/2026

Date: 15th Mar - 30th Apr

Individual Component: Risk and Compliance Audit: Implementing Attacks 1 to 4 covering certificate integrity verification, privacy budget audit, reconstruction attack, and membership inference attack.

Summary of RP work carried out during the reporting period

Date	RP Work
15/03/2026	Reviewed k-anonymity definitions and quasi-identifier concepts. Designed the Attack 5 metadata leakage audit approach using five quasi-identifiers.
17/03/2026	Studied cosine similarity re-identification methods and embedding design for Attack 6. Completed the full attack planning document for all six attacks.
19/03/2026	Started implementing Attack 1: certificate integrity verification using HMAC-SHA256. Wrote the HMAC recomputation and comparison logic.
21/03/2026	Added environment variable injection for the HMAC key in Attack 1. Verified the key is never hardcoded in the source code. Tested against a sample certificate file.
23/03/2026	Ran Attack 1 against all pipeline certificate files. All signatures verified correctly. Attack 1 PASS result confirmed and documented.
25/03/2026	Began implementing Attack 2: privacy budget audit. Wrote the parser to read the Opacus calibration log files and extract epsilon values.
27/03/2026	Completed epsilon comparison logic for Attack 2. Tested against Config A, B and C epsilon values. All three matched within the 0.0001 tolerance. PASS confirmed.
01/04/2026	Started implementing Attack 3: Laplace reconstruction attack. Implemented the Laplace bound formula: expected error = 2 x sensitivity / epsilon.
03/04/2026	Ran Monte Carlo simulation with 10000 Laplace noise samples to validate the formula implementation before applying it to the real calibration statistics.
07/04/2026	Attack 3 results: observed error 0.6246, expected bound 0.6667, ratio 0.94. PASS confirmed. Documented results and generated the attack report.
10/04/2026	Began implementing Attack 4 membership inference. Built the shadow classifier training pipeline using scikit-learn following the Shokri et al. protocol.
14/04/2026	Ran Attack 4 against all four DP configurations. AUC results: strong 0.0093, medium 0.2387, weak 0.4645, no-DP 0.0063. All below 0.55 threshold. PASS confirmed.
17/04/2026	Started writing individual attack report JSON files and PDF summaries for Attacks 1 to 4. Reviewed formatting with supervisor.
21/04/2026	Applied supervisor feedback to all four attack reports. Confirmed consistent formatting and content across all reports. Phase complete for Attacks 1 to 4.

Supervisor's comments.

Student progressing well in research project: Yes / No

If NO:

Project work to be carried out before next evaluation period:

Signature:

Date: 22/04/2026

Date: 1st May - 20th Jun

Individual Component: Risk and Compliance Audit: Implementing Attacks 5 and 6, building the compliance certificate generator, and running the full attack suite end-to-end.

Summary of RP work carried out during the reporting period

Date	RP Work
01/05/2026	Started Attack 5: metadata leakage audit. Designed the PII regex pattern set covering 12 identifier types including GPS coordinates, names, IDs and emails.
05/05/2026	Implemented the PII scanner for CSV column names and row values across all five town metadata files from the CARLA simulator output.
07/05/2026	Ran PII scan across town01 to town05 (6000 rows each, 10015 total files). Zero PII findings returned. Documented result and prepared justification.
09/05/2026	Implemented k-anonymity check using five quasi-identifiers: Weather Category, Time Of Day, Weather Label, Event Type and Town across 10000 rows.
12/05/2026	K-anonymity result: k=1 found, meaning 19 unique frame combinations exist. Declared as a low-severity weakness in the Attack 5 report. Verdict: REVIEW.
14/05/2026	Completed Attack 5 report. Documented full justification for zero PII findings: CARLA is a simulator and does not generate real personal data by design.
17/05/2026	Began Attack 6: re-identification attack. Designed a 12-dimensional embedding using percentile rank of Num_Objects and one-hot encoding of categorical fields.
19/05/2026	Implemented embedding generation for LIDAR and trajectory metadata. Tested on town01 sample rows to verify the embedding output structure.
21/05/2026	Ran cosine similarity comparison: 1000 real vs 1500 synthetic embeddings for LIDAR. Re-identification rate 0%, max similarity 0.9568. PASS confirmed.
23/05/2026	Trajectory re-identification rate 0%, max similarity 0.9894. PASS confirmed. Camera modality skipped as images are AES-256-GCM encrypted.
26/05/2026	Validated Attack 6 threshold by injecting five real rows into the synthetic set. All five were flagged at similarity 1.0, confirming the 0.999 threshold is correct.
28/05/2026	Implemented the compliance certificate generator. Added SHA-256 fingerprint across all six combined JSON reports and HMAC-SHA256 signing.
02/06/2026	Generated final compliance certificate. Certificate ID: R26-RC-760AD35A004D6A. Overall verdict: PASS with REVIEW note for k=1 weakness in Attack 5.
10/06/2026	Ran full attack suite end-to-end using run_all_attacks.py. Identified and fixed HMAC key injection issue - subprocess was not receiving the key from environment.
15/06/2026	Fixed ReportLab colour crash in certificate generator and verdict extraction logic in run_all_attacks.py. Re-ran full suite. All six attacks successful. Certificate generated correctly.

Supervisor's comments.

Student progressing well in research project: Yes / No

If NO:

Project work to be carried out before next evaluation period:

Signature:

Date: 16/06/2026

Date: 21st Jun - 31st Jul

Individual Component: Risk and Compliance Audit: Writing the Risk and Compliance chapter of the research report covering GDPR background, all six attack methodologies, results and the compliance certificate.

Summary of RP work carried out during the reporting period

Date	RP Work
21/06/2026	Started Phase 4 report writing. Outlined the full Risk and Compliance chapter structure and confirmed section headings with the supervisor.
24/06/2026	Wrote the GDPR compliance background section covering Articles 5, 25 and 32 in the context of the AV data pipeline design.
26/06/2026	Wrote the Attack 1 and Attack 2 methodology sections with code-level explanations of the HMAC verification and epsilon audit approaches.
28/06/2026	Wrote the Attack 3 and Attack 4 methodology sections. Included the Laplace bound derivation and the shadow classifier design description.
01/07/2026	Wrote the Attack 5 methodology and results sections. Integrated the full PP2 justification argument for why zero PII findings is the correct expected result.
03/07/2026	Wrote the Attack 6 methodology section. Explained the embedding design, threshold selection rationale and the injection-based validation approach.
07/07/2026	Completed first draft of the full Risk and Compliance chapter and submitted for group review.
10/07/2026	Received group feedback. Revised the Attack 5 results section to add a clearer explanation of the k-anonymity finding and its GDPR implications.
13/07/2026	Revised the certificate generation section to better explain the distinct roles of SHA-256 fingerprinting and HMAC signing in the certificate design.
16/07/2026	Integrated the Risk and Compliance chapter with the main group report draft. Checked all attack result values are consistent throughout the combined document.
20/07/2026	Made final edits to the chapter based on the combined report review findings. Prepared the compliance certificate appendix and attack report summaries.
24/07/2026	Ran the full attack suite one final time to confirm all results are reproducible and consistent with the values reported in the submitted chapter.
28/07/2026	Submitted final Risk and Compliance component deliverables to the group repository. All six attack reports and the compliance certificate included.

Supervisor's comments.

Student progressing well in research project: Yes / No

If NO:

Project work to be carried out before next evaluation period:

Signature: 

Date: 29/07/2026

Date: 1st Aug - 15th Oct

Individual Component: Risk and Compliance Audit: Group final report integration, post-submission review, project closure and final evaluation of all deliverables.

Summary of RP work carried out during the reporting period

Date	RP Work
03/08/2026	Group final report integration completed. Submitted combined final report draft for supervisor review.
07/08/2026	Received supervisor review comments. Made minor edits to the certificate section as requested and resubmitted for final approval.
11/08/2026	Final report approved by supervisor. All deliverables for the Risk and Compliance component confirmed complete and submitted.
15/08/2026	Post-submission review. Documented lessons learned from the attack implementation and GDPR compliance work throughout the project.
19/08/2026	Reviewed all six attack scripts and verified all code comments are complete and accurate for the final archived version.
24/08/2026	Confirmed run_all_attacks.py is fully self-contained and reproducible from a clean environment. Archived all scripts and outputs.
02/09/2026	Reviewed the complete project timeline. Confirmed all deliverables are documented correctly in the log and all submissions are in order.
08/09/2026	Re-read the full Risk and Compliance chapter and all six attack reports to refresh understanding ahead of final evaluation.
12/09/2026	Reviewed all six attacks in detail covering HMAC key injection, epsilon matching, Laplace bound, shadow classifier, PII scan and cosine similarity embedding.
17/09/2026	Reviewed compliance certificate structure including SHA-256 fingerprint computation, HMAC signing and certificate ID format.
22/09/2026	Final check of all numeric results across all six attacks for consistency with the submitted report. Confirmed all values are correct.
26/09/2026	Submitted final log book and all supporting documentation. Project preparation confirmed complete.
07/10/2026	Attended viva examination for R26-CS-010 Risk and Compliance component.
11/10/2026	Completed all post-viva follow-up actions. Submitted final documentation to supervisor.
15/10/2026	Project R26-CS-010 Risk and Compliance component officially concluded. All materials archived.

Supervisor's comments.

Student progressing well in research project: Yes / No

If NO:

Project work to be carried out before next evaluation period:

Signature:

Date: 16/10/2026